

Experiment Report: Retail Expansion

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Abstract

This report presents a data-driven framework for retail expansion site selection, integrating multiple linear regression, K-means clustering, discounted cash flow (DCF) analysis, Monte Carlo simulation, and a comparative experiment between independent project NPV and enterprise value increment. Using synthetic data representing 184 geographic areas with six demographic and competitive features, the analysis identifies key factor of sales, segments regions into three potential tiers, quantifies investment risk through simulation, and reveals the systematic overestimation of independent NPV when internal market competition is present. The results show that while population and income positively influence sales, competitor density has a negative effect. The independent NPV sum of selected new stores is approximately \$370,000, but after accounting for a 20% sales erosion on existing stores, the true enterprise value increment reduces to roughly \$134,000—a 63% value overstatement. The study concludes that a holistic enterprise-value perspective is essential for sound capital budgeting decisions in multi-store retail networks.

1 Introduction

Retail expansion decisions involve substantial capital investment and are subject to demand uncertainty, cost variations, and competitive pressures. Traditional qualitative measure often fail to quantify regional potential and risk systematically. This experiment develops a comprehensive, data-driven methodology that moves from demand forecasting and regional segmentation to financial evaluation and risk simulation, ultimately providing defensible site recommendations.

The analysis is conducted on a synthetic random dataset of 184 areas, each characterized by annual sales (*total_sales*) and six explanatory variables: total population, median household income, median age, education percentage, competitor count, and transit convenience. The report details the entire analytical pipeline and concludes with a

critical comparison between independent project NPV and enterprise value increment, highlighting the importance of considering internal market competition.

2 Methodology

2.1 Data Generation and Preparation

The dataset is generated synthetically to mimic realistic retail market conditions. For each of 184 zip-codes, order-level sales and profit data are simulated (800–2,000 orders per area), then aggregated to annual area-level totals and average profit margins. Demographic features are drawn from normal, beta, and Poisson distributions with parameters chosen to reflect plausible real-world ranges. Transit convenience is derived as the inverse of distance to transit.

All numerical variables are standardized using z-score normalization to eliminate scale effects prior to regression and clustering:

$$z_j = \frac{x_j - \mu_j}{\sigma_j} \quad (1)$$

2.2 Multiple Linear Regression

A linear regression model is fitted to predict total sales from the six standardized features:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \varepsilon \quad (2)$$

Coefficients are estimated via ordinary least squares. Model diagnostics include coefficient interpretation, residual distribution, and Q-Q plot to assess normality of errors.

2.3 K-Means Clustering

To segment areas into distinct potential tiers, K-means clustering is applied to the standardized feature matrix. The optimal number of clusters is selected by balancing the silhouette score and business interpretability. Although the silhouette score peaks at $K = 7$, a $K = 3$ solution is preferred based on the principle of parsimony (negligible improvement in silhouette score from $K = 3$ to $K = 7$) and decision-making clarity (three intuitive tiers: High, Median, and Low Potential). The cluster centroids are back-transformed to original units for interpretation.

2.4 Deterministic DCF and NPV

For each representative area from each cluster, a deterministic net present value is calculated under the following assumptions:

- Initial investment $I_0 = \$500,000$
- Forecast horizon $T = 5$ years
- Annual sales growth $g = 5\%$
- Operating cost ratio $c = 65\%$
- Discount rate $r = 10\%$

The net cash flow in year t is:

$$NCF_t = S_0(1 + g)^t(1 - c) \quad (3)$$

Then NPV is:

$$NPV = -I_0 + \sum_{t=1}^T \frac{NCF_t}{(1 + r)^t} \quad (4)$$

2.5 Monte Carlo Simulation

To rigorously quantify the uncertainty in NPV arising from stochastic economic conditions, we employ Monte Carlo simulation. The model treats annual revenue growth and operating cost ratios as random variables, capturing the volatility inherent in retail operations.

2.5.1 Distributional Assumptions and Parameter Selection

The key uncertain parameters are the annual sales growth factor G_t and the operating cost ratio C .

1. **Annual Sales Growth Factor (G_t):** We assume that the growth factor $G_t = 1 + g_t$ (where g_t is the growth rate) follows a **Lognormal distribution**. This ensures that sales never decline below zero and captures the right-skewed nature of positive growth shocks. The probability density function is:

$$f(G) = \frac{1}{G\sigma_g\sqrt{2\pi}} \exp\left(-\frac{(\ln G - \mu_g)^2}{2\sigma_g^2}\right), \quad G > 0 \quad (5)$$

Based on historical U.S. retail sales trends, we set the expected annual growth factor $\mathbb{E}[G_t] = 1.05$ (i.e., 5% growth) and a standard deviation $\sigma[G_t] = 0.03$. Given the properties of the lognormal distribution, the parameters μ_g and σ_g are derived via

moment matching:

$$\sigma_g = \sqrt{\ln \left(1 + \frac{\sigma[G_t]^2}{\mathbb{E}[G_t]^2} \right)} = \sqrt{\ln \left(1 + \frac{0.03^2}{1.05^2} \right)} \approx 0.0285 \quad (6)$$

$$\mu_g = \ln(\mathbb{E}[G_t]) - \frac{1}{2}\sigma_g^2 = \ln(1.05) - \frac{1}{2}(0.0285)^2 \approx 0.0484 \quad (7)$$

2. **Operating Cost Ratio (C):** We assume that the cost ratio C follows a **Beta distribution**, which is naturally bounded between 0 and 1, making it ideal for modeling cost margins. The probability density function is:

$$f(c) = \frac{c^{\alpha-1}(1-c)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 < c < 1 \quad (8)$$

where $B(\alpha, \beta)$ is the Beta function. We set the expected cost ratio $\mathbb{E}[C] = 0.65$ and standard deviation $\sigma[C] = 0.05$. Using the method of moments, the shape parameters α and β are solved as:

$$\alpha = \mathbb{E}[C] \left(\frac{\mathbb{E}[C](1 - \mathbb{E}[C])}{\sigma[C]^2} - 1 \right) = 0.65 \left(\frac{0.65 \times 0.35}{0.05^2} - 1 \right) \approx 58.5 \quad (9)$$

$$\beta = (1 - \mathbb{E}[C]) \left(\frac{\mathbb{E}[C](1 - \mathbb{E}[C])}{\sigma[C]^2} - 1 \right) = 0.35 \left(\frac{0.65 \times 0.35}{0.05^2} - 1 \right) \approx 31.5 \quad (10)$$

2.5.2 Simulation Algorithm

For each candidate region with a base annual revenue S_0 , we perform $M = 5,000$ simulation iterations. The computational steps for the i -th iteration are as follows:

1. **Sampling:** Draw 5 independent growth factors $\mathbf{G}^{(i)} = \{G_1^{(i)}, \dots, G_5^{(i)}\}$ from the specified Lognormal distribution, and draw one cost ratio $c^{(i)}$ from the Beta distribution.
2. **Cash Flow Projection:** For each year $t \in \{1, \dots, 5\}$, calculate the annual sales and net cash flow:

$$S_t^{(i)} = S_0 \cdot \prod_{k=1}^t G_k^{(i)} \quad (11)$$

$$NCF_t^{(i)} = S_t^{(i)} \cdot (1 - c^{(i)}) \quad (12)$$

3. **NPV Calculation:** Compute the net present value for this specific scenario:

$$NPV^{(i)} = -I_0 + \sum_{t=1}^5 \frac{NCF_t^{(i)}}{(1+r)^t} \quad (13)$$

where $I_0 = \$500,000$ and the discount rate $r = 10\%$.

2.5.3 Statistical Inference and Risk Metrics

After completing M iterations, the empirical distribution of NPV is analyzed. By the **Weak Law of Large Numbers (WLLN)**, the sample mean converges to the expected value. By the **Central Limit Theorem (CLT)**, the sample mean asymptotically follows a normal distribution, allowing for the construction of confidence intervals.

We derive the following key risk metrics for decision-making:

- **Expected NPV:** $\mathbb{E}[NPV] \approx N\bar{P}V = \frac{1}{M} \sum_{i=1}^M NPV^{(i)}$
- **Volatility (Standard Deviation):** $\sigma_{NPV} = \sqrt{\frac{1}{M-1} \sum_{i=1}^M (NPV^{(i)} - N\bar{P}V)^2}$
- **Probability of Loss:** $P(\text{Loss}) \approx \frac{1}{M} \sum_{i=1}^M \mathbb{I}\{NPV^{(i)} < 0\}$ (where $\mathbb{I}$ is the indicator function)
- **Value at Risk (VaR) at 5%:** The 5th percentile of the sorted NPV sequence, representing the minimum value expected with 95% confidence.

2.6 Enterprise Value Increment Experiment

Discounted Cash Flow Model is based on an ideal assumption that the cash flow of new stores is independent of the company's existing business. In reality, newly opened stores may compete with existing stores (market erosion), reduce procurement costs through economies of scale (synergy), and may face overall financial constraints for the company. A comparative experiment is designed to contrast the independent project NPV approach with an enterprise-value perspective. This new experiment assumes:

- The firm currently operates 5 existing mature stores, each with annual sales around \$350,000–\$450,000 and a 3% growth rate.
- New stores, when opened, cause a permanent 20% sales erosion on the nearest existing store.
- The benchmark enterprise value (EV_{base}) is the present value of cash flows from existing stores only.
- The expanded enterprise value (EV_{exp}) includes the cash flows of existing stores after erosion plus the cash flows of new stores, minus the total investment.
- The true value created by expansion is $\Delta EV = EV_{exp} - EV_{base}$.

This is compared to the sum of independent NPVs for the selected new stores.

3 Results

3.1 Regression Analysis

The standardized regression coefficients reveal the following relationships:

Table 1: Standardized Regression Coefficients

Variable	Coefficient
total_population	+0.45
median_household_income	+0.32
median_age	+0.18
education_pct	+0.09
competitor_count	-0.27
transit_convenience	+0.12

Population and income are the strongest positive drivers, while competitor count exerts a notable negative influence. The residuals are approximately normally distributed, as confirmed by the histogram and Q-Q plot, indicating that the linear model provides a reasonable fit.

3.2 Clustering and Segmentation

The silhouette scores for $K = 2$ to $K = 7$ are:

Table 2: Silhouette Scores for Different K Values

K	Silhouette Score
2	0.1260
3	0.1321
4	0.1345
5	0.1280
6	0.1340
7	0.1347

Although $K = 7$ yields the highest score, the marginal improvement over $K = 3$ (0.0026) is negligible, while the number of clusters more than doubles. Consequently, $K = 3$ is selected, resulting in three potential tiers: High, Median, and Low Potential.

3.3 Deterministic NPV

For the three randomly selected representative areas, the deterministic NPVs are:

Only the High and Low potential areas have positive NPV, while the Median potential area is negative and would be rejected under independent project criteria.

Table 3: Deterministic NPV for Representative Areas

Zip Code	Potential	Deterministic NPV
ZIP10123	High	+\$223,876
ZIP10048	Median	-\$116,678
ZIP10031	Low	+\$146,026

3.4 Monte Carlo Simulation Results

The simulation results for the three candidates are summarized as follows:

Table 4: Monte Carlo Simulation Risk Metrics

Zip Code	Potential	E[NPV]	Std[NPV]	P(Loss)	VaR_5%	Score
ZIP10123	High	\$259,276	\$113,116	0.00%	\$79,327	0.70
ZIP10031	Low	\$179,713	\$99,671	2.0%	\$21,532	0.66
ZIP10048	Median	-\$97,724	\$59,290	94%	-\$192,184	0.31

The High Potential area dominates on all metrics: highest expected NPV, lowest volatility, near-zero loss probability, and highest composite score (0.70). The Low Potential area also shows positive expected NPV but with higher risk. The Median area is clearly unattractive.

3.5 Enterprise Value Increment Experiment

The baseline enterprise value from the five existing stores is:

$$EV_{base} = \$2,936,362 \quad (14)$$

After incorporating the two positive-NPV new stores (ZIP10123 and ZIP10031) and applying a 20% erosion to the nearest existing store for each, the expanded enterprise value is:

$$EV_{exp} = \$3,070,414 \quad (15)$$

Thus, the true value increment is:

$$\Delta EV = \$134,052 \quad (16)$$

In contrast, the sum of independent NPVs for the two new stores is:

$$\sum NPV_{independent} = \$369,902 \quad (17)$$

The overstatement amounts to:

$$\$369,902 - \$134,052 = \$235,850 \quad (18)$$

This represents a 63% overestimation of the value created by the expansion.

Table 5: Comparison of Independent NPV and Enterprise Value Increment

Method	Value (\$)	Decision
Independent NPV sum	369,902	Expand (all positive)
Enterprise Value Increment ΔEV	134,052	Expand

Both methods support expansion, but the independent NPV approach materially overstates the net benefit.

4 Discussion

The results underscore several important insights for retail site selection:

1. **Key Drivers:** Population and income are the primary positive determinants of sales, while competition reduces sales significantly. This aligns with retail location theory and suggests that expansion should prioritize densely populated, high-income areas with low competitor density.
2. **Segmentation for Decision-Making:** The $K = 3$ clustering solution provides a simple yet effective classification. The slight improvement in silhouette score at $K = 7$ does not justify the added complexity; three tiers are sufficient for strategic portfolio management.
3. **Risk Quantification:** Monte Carlo simulation transforms a single-point NPV estimate into a distribution of possible outcomes. The High Potential area not only has the highest expected NPV but also the lowest risk of loss, making it the most robust choice. The composite score, integrating expected return, volatility, and loss probability, offers a transparent ranking criterion.
4. **The Internal Market Competition Effect:** The comparative experiment reveals that independent NPV can be dangerously misleading when new stores draw sales from existing outlets. The 20% erosion assumption, while stylized, illustrates a common real-world phenomenon. In this case, nearly two-thirds of the projected independent value is actually a transfer from existing stores, not new wealth creation. This has profound implications for capital budgeting in multi-unit retail chains.

5. **Practical Implications:** Independent NPV remains a useful initial screening tool. However, final go/no-go decisions should be based on enterprise-wide incremental value, explicitly modeling cannibalization, synergies, and network effects. The enterprise value approach provides a more conservative and economically sound measure of shareholder value creation.
6. **Limitations:** The analysis relies on synthetic data, so the specific numerical results are illustrative rather than prescriptive. The erosion rate is assumed constant and applied only to the nearest store; in reality, cannibalization patterns are more complex. Future work could incorporate spatial econometric models or Bayesian parameter updates to improve robustness.

5 Implementation Challenges and Corrections

During the implementation of the analytical pipeline, several technical and methodological challenges were encountered. The key issues, their root causes, and the corresponding corrective actions are summarized below.

- **Issue: Negative values generated for population and income.**
Cause: The normal distribution was applied without truncation.
Resolution: Applied `.clip(0)` to enforce non-negativity.
- **Issue: Raw data was overwritten after standardization.**
Cause: Direct assignment `df[features] = X_scaled` replaced the original columns.
Resolution: Preserved the raw data in `df_features_raw` and stored standardized values in a separate table `df_features_scaled`.
- **Issue: The `np.npv` function raised an error.**
Cause: The function was removed from NumPy in version 1.20+.
Resolution: Installed the `numpy-financial` library and used `npf.npv` instead.
- **Issue: Explosive growth rates occurred in the Monte Carlo simulation (up to 105%).**
Cause: The cumulative product was incorrectly applied as `cumprod(1+growth)`, while the growth factors already included the base of 1.
Resolution: Corrected the implementation to `cumprod(growth_factors)`.
- **Issue: NPV boxplots displayed values in scientific notation.**
Cause: Large data magnitudes automatically triggered scientific formatting.
Resolution: Changed the origin data of sales amount and value per sale.

- **Issue: Chinese characters appeared as blank squares in plots.**
Cause: Missing font configurations for Chinese characters in the plotting environment.
Resolution: Replaced all chart labels and annotations with English text.
- **Issue: Silhouette scores were low, and the optimal K was automatically selected as 7.**
Cause: The synthetic data lacked a strong natural clustering structure.
Resolution: Combined the elbow method with business requirements to select $K = 3$, and justified this choice explicitly in the report.
- **Issue: The probability of loss ($P(\text{Loss})$) was consistently calculated as zero.**
Cause: The simulated sales magnitudes were relatively high compared to the investment, and the volatility parameters were conservative.
Resolution: Acknowledged this as a high safety margin in the report and recommended sensitivity analysis to reveal the threshold for risk emergence.
- **Issue: The VaR for the Low Potential region outperformed that of the Median Potential region.**
Cause: Random sampling selected an atypical store with high sales within the low-potential cluster.
Resolution: Explicitly discussed this phenomenon in the analysis, highlighting that cluster labels do not perfectly align with financial metrics, which enriched the interpretative depth of the report.
- **Issue: The independent NPV sum included stores with negative NPV in the enterprise value increment comparison.**
Cause: The sum was performed without filtering out projects with $NPV < 0$.
Resolution: Restricted the comparison to only the subset of stores with positive independent NPV.

6 Conclusion

This report presents a complete, reproducible framework for retail expansion site selection, integrating regression, clustering, DCF, Monte Carlo simulation, and an enterprise-value incremental analysis. The main conclusions are:

- Population and income are the dominant positive factors; competition is the main negative factor.

- A three-tier clustering (High/Median/Low Potential) balances statistical adequacy with business interpretability.
- Monte Carlo simulation effectively quantifies risk and supports the selection of the High Potential area as the preferred candidate.
- The independent NPV approach overstates the true value of expansion by approximately 63% when cannibalization is ignored, highlighting the necessity of a holistic enterprise-value perspective.

In practice, retailers should use independent NPV for initial screening but must adopt an enterprise-wide incremental cash flow approach for final investment decisions. The framework demonstrated here can be adapted with real data and refined with more sophisticated spatial and dynamic models, ultimately leading to more informed and defensible expansion strategies.

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